



Operating Systems & Networks

CTEC1704C

Lecture - 4 : CPU

Dr. Sameer Jha, SFHEA
Faculty of Computing
De Montfort University Cambodia



Outline the topics to be covered

This lecture introduces the hardware foundations needed to understand operating systems and endpoint security.

- Technology trends in computer hardware
- Moore's Law and performance growth
- Core components of a computer system
- Instruction sets: how software talks to hardware



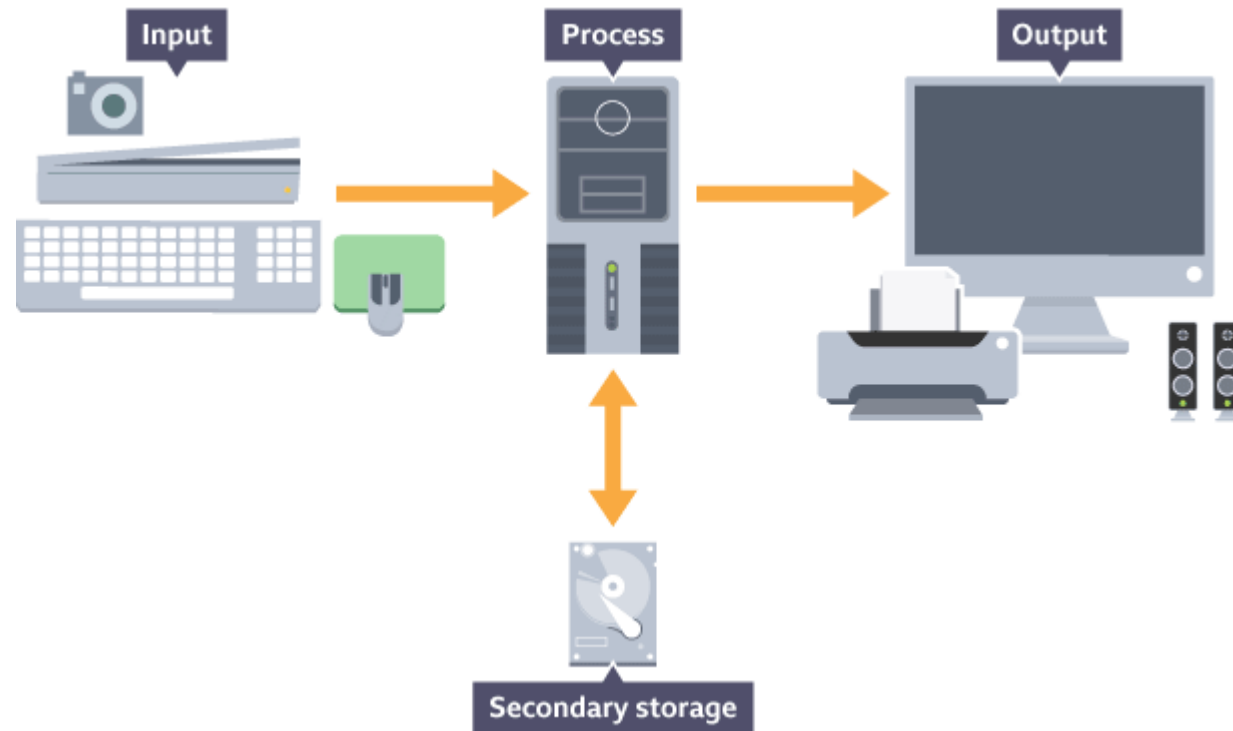
**Computer Organization
and Design**

Patterson, Hennessy
5/E Morgan Kaufmann
2013
ISBN: 0124077269



Hardware

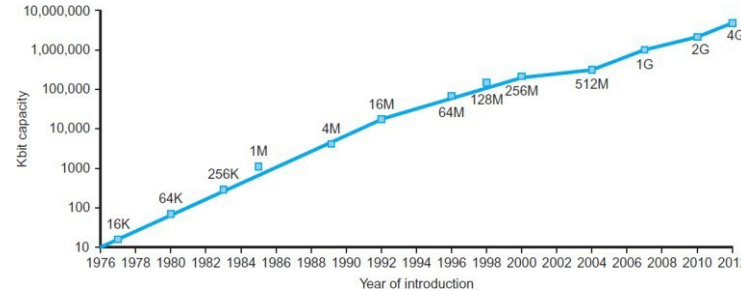
Hardware is the physical components of the computer, such as the **central processing unit (CPU)**, **hard disk**, **monitor**, keyboard and mouse.





Technology Trends

- Electronics technology continues to evolve
 - Increased capacity and performance
 - Reduced cost



Year	Technology	Relative performance/cost
1951	Vacuum tube	1
1965	Transistor	35
1975	Integrated circuit (IC)	900
1995	Very large scale IC (VLSI)	2,400,000
2013	Ultra large scale IC	250,000,000,000

Key takeaway: performance has increased exponentially while cost has decreased.



Components of a Computer

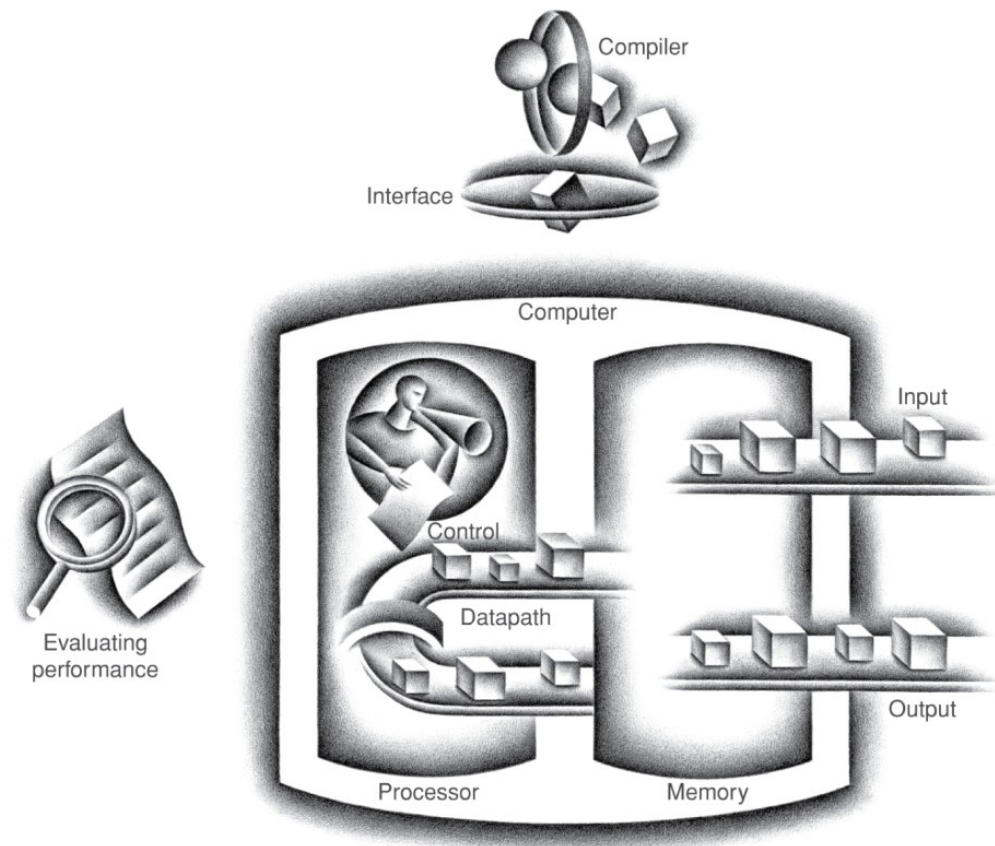
- **Same components**

for all kinds of computer

- Desktop, server, embedded, mobile, etc.

- **Input/output includes**

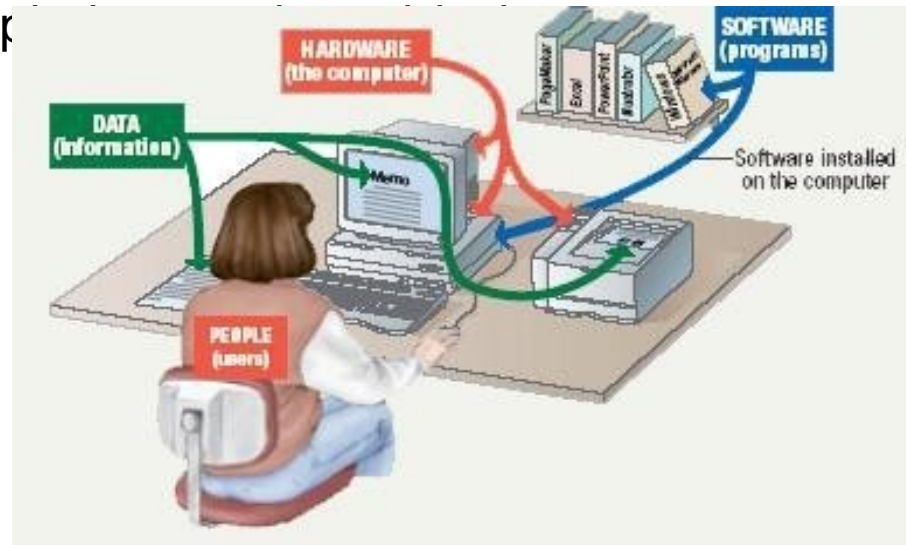
- User-interface devices
 - Display, keyboard, mouse
- Storage devices
 - Hard disk, CD/DVD, flash
- Network adapters
 - For communicating with other computers



Looking Inside the Computer

Computer systems have four parts

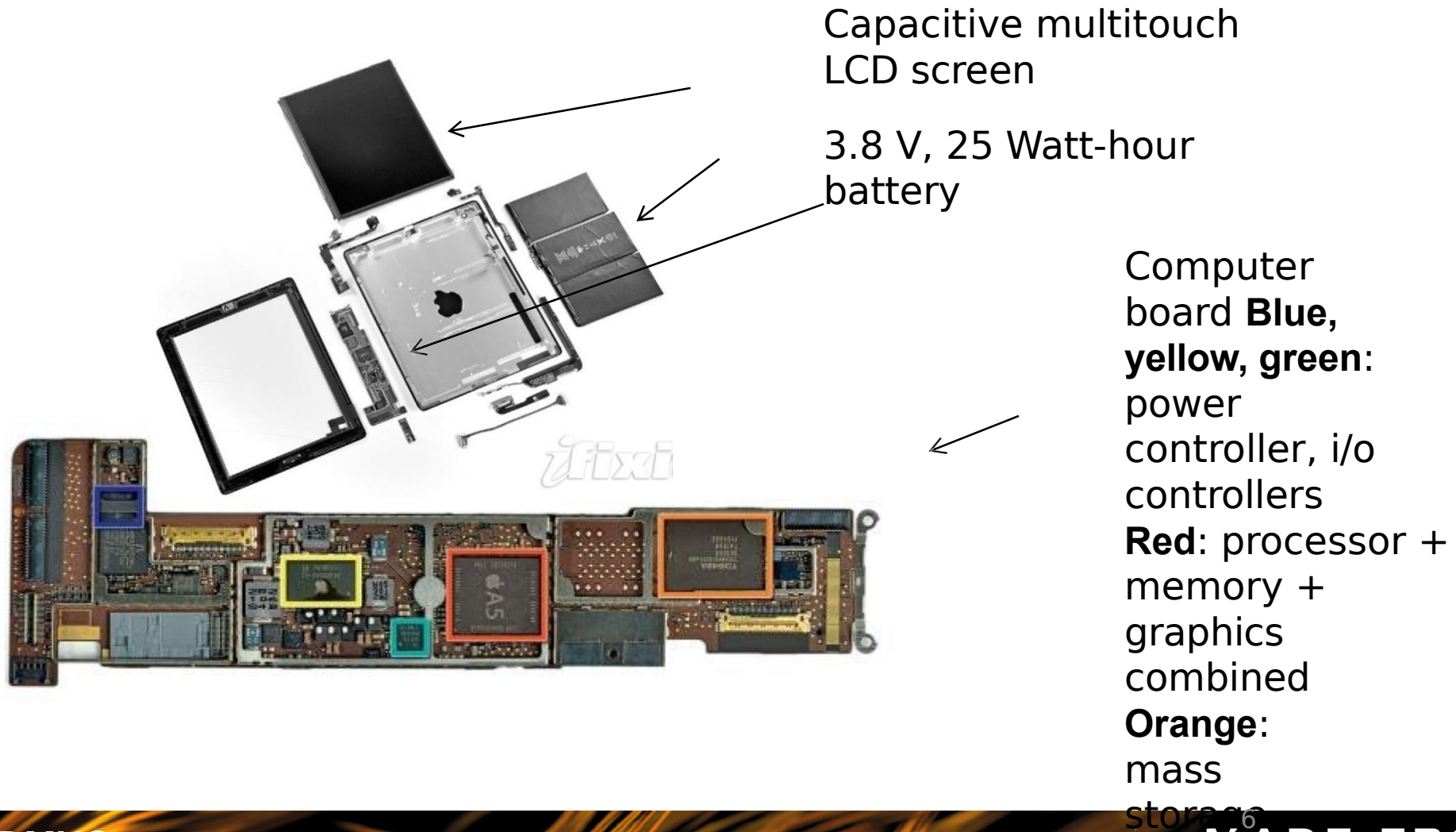
- **Hardware**: physical components (CPU, RAM, storage, I/O)
- **Software**: system software, applications, middleware
- **Data**: information processed by the system
- **User**: people



All four must work together for a system to function reliably and securely

Opening the Box

Modern devices integrate multiple components onto a single board to improve performance and power efficiency.





Inside the Processor (CPU)

Datapath: performs operations on data

Control: sequences datapath, memory, ...

Cache memory

Small fast SRAM memory for immediate access to data

Inside the Processor

Modern CPUs come in a variety of core configurations. Here are some common ones you might see:

1. Single-Core

Older or very low-power devices sometimes have only one CPU core. Can only process one set of instructions (thread) at a time.

2. Dual-Core

Two cores in a single CPU package. Good balance for budget desktops, laptops, or low-power devices.

3. Quad-Core

Four cores. Common in many modern desktops and laptops. Offers better multitasking and performance than dual-core.

4. Hexa-Core

Six cores. Provides a performance boost for tasks like video editing, 3D rendering, or gaming.

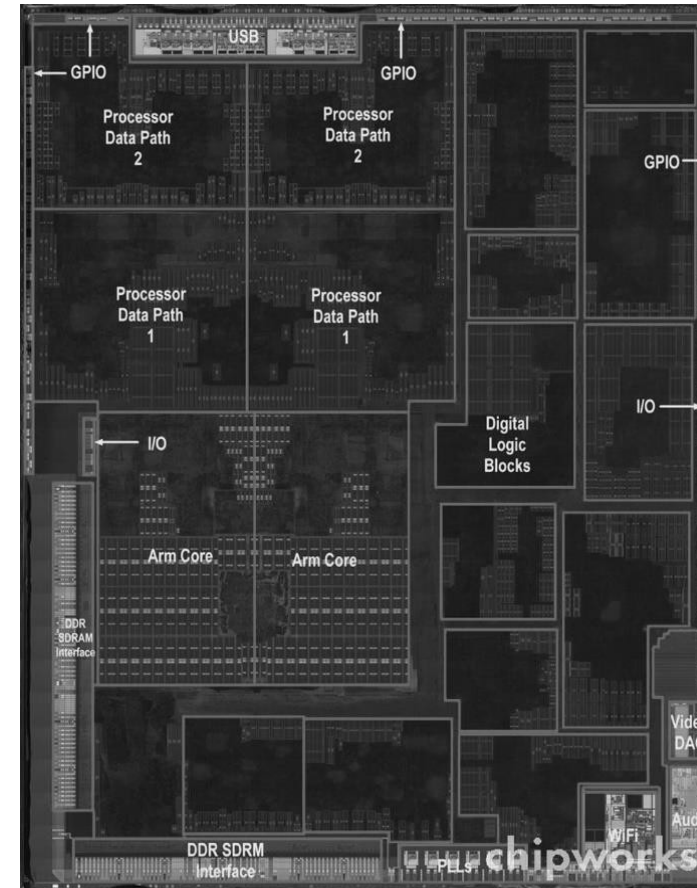
5. Octa-Core

Eight cores. Often found in higher-end desktops and smartphones. Good for heavily multi-threaded tasks.

6. Deca-Core (and higher)

Ten or more cores (e.g., 10, 12, 16, etc.). Common in high-performance workstation or server CPUs. Can handle many simultaneous threads.

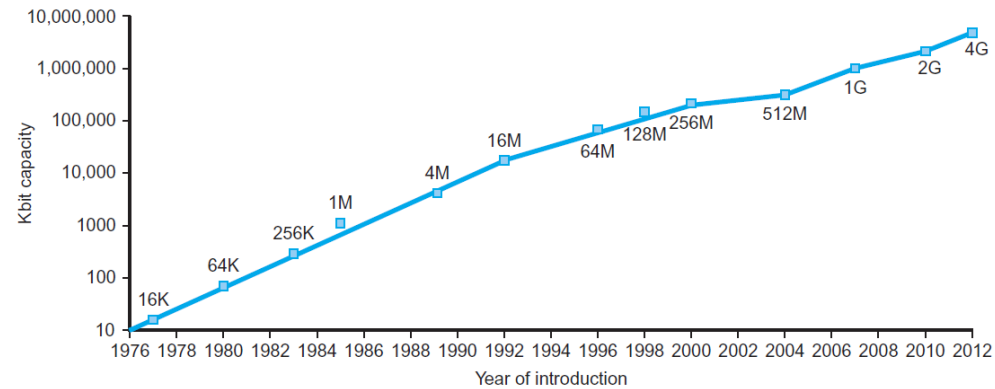
12.1mm



- Apple A5

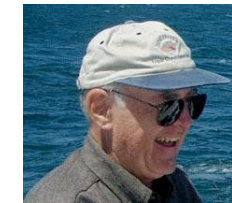
More cores allow better multitasking and parallel processing.

- Electronics technology continues to evolve
 - Increased capacity and performance
 - Reduced cost



DRAM capacity

Year	Technology	Relative performance/cost
1951	Vacuum tube	1
1965	Transistor	35
1975	Integrated circuit (IC)	900
1995	Very large scale IC (VLSI)	2,400,000
2013	Ultra large scale IC	250,000,000,000



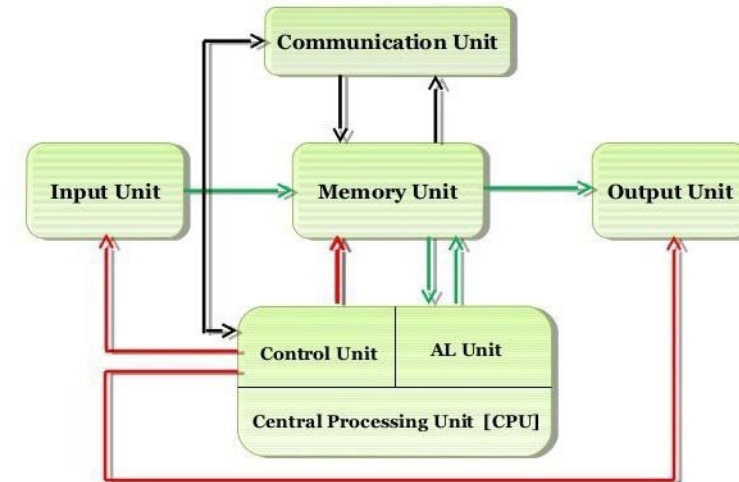
- # 2025-26 DMUC

Components of a Computer

The core components remain consistent across all types of computers, including desktops, servers, embedded systems, and mobile devices.

- 1-a Central Processing Unit (CPU)
- 2-Primary Storage or Random Access Memory (RAM)
- 3-Secondary Storage
- 4-Input/Output devices (e.g., screen, keyboard, mouse)
- 5-interconnection referred to as the Bus.

Basic Components of Computer



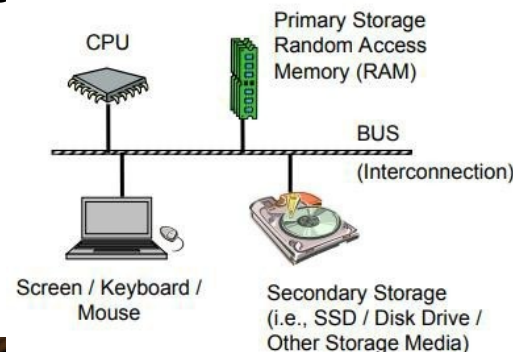
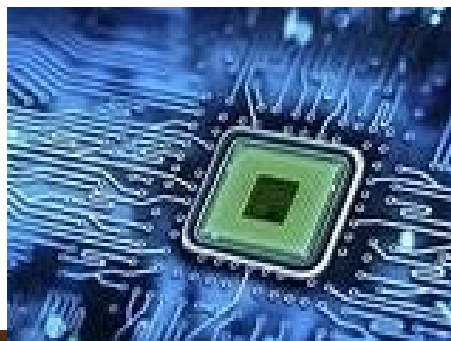
The architecture is typically referred to as the Von Neumann Architecture and was described in 1945 by the mathematician and physicist John von Neumann.

This design influences how attacks

Semiconductor Technology

Semiconductors are the foundation of all modern digital electronics. Key parts made with semiconductors include:

- **Central Processing Unit (CPU)**: The brain of the computer, responsible for executing instructions and performing calculations. Made with billions of semiconductor transistors.
- **Memory (RAM and ROM)**: Random Access Memory (RAM) and Read-Only Memory (ROM) chips store data temporarily and permanently.
- **Graphics Processing Unit (GPU)**: Specialised for rendering images, videos, and performing parallel computations.
- **Solid-State Drives (SSDs)**: Use NAND flash memory, which is a semiconductor-based technology, for fast storage and retrieval of data.
- **Motherboard Chipsets**: Components like the Northbridge, Southbridge, and other controllers on the motherboard are made with semiconductors to manage communication between different hardware parts.





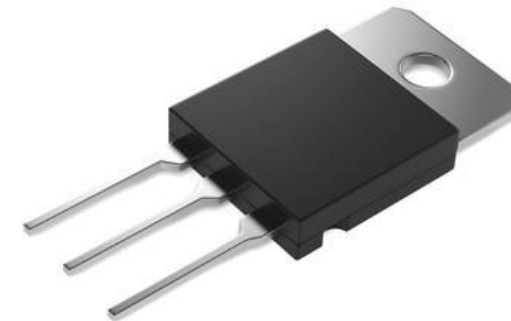
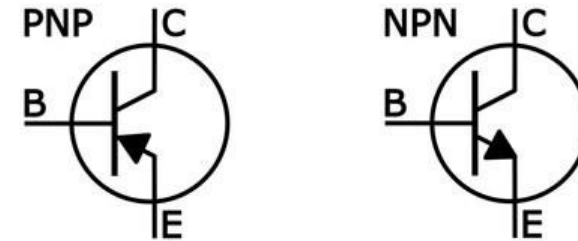
Transistor: on/off switch

The transistor is a semiconductor device that functions as an **on/off switch**. It is a fundamental building block of modern electronics, including computers.

Here's how it works:

- **On/Off Switch:** In digital circuits, transistors control the flow of electrical current by switching it on or off, corresponding to binary states (1 and 0).
- **Amplification:** Transistors can also amplify signals, which is crucial in analogue devices audio equipment.

Integrated circuits combine many transistors on a single chip



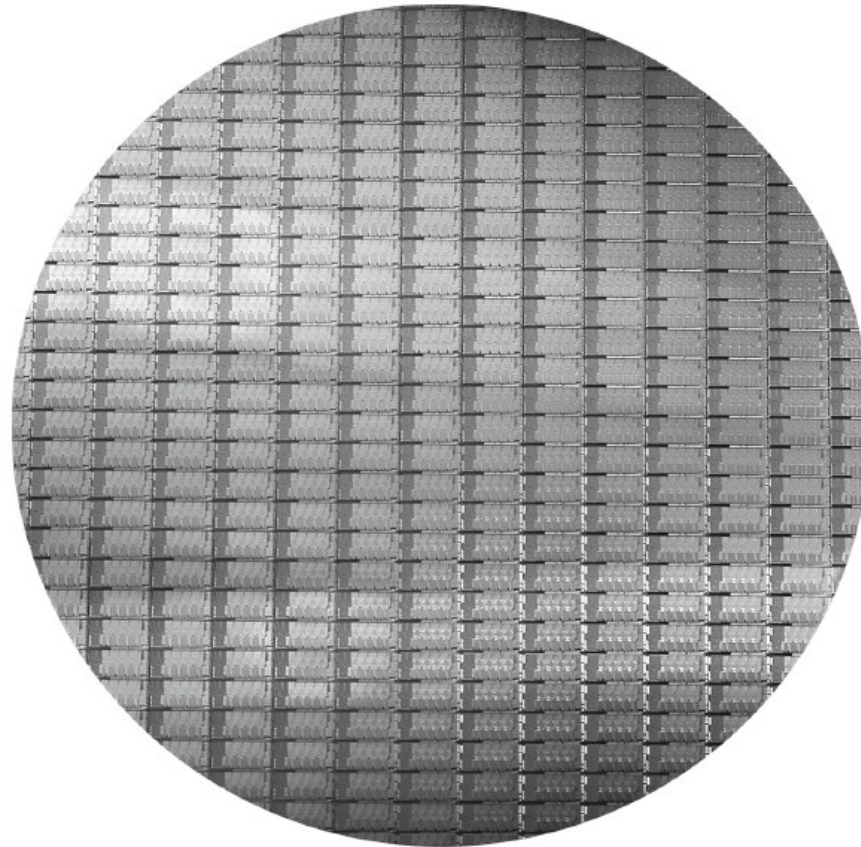


Semiconductor Technology

- Transistor: on/off switch
- Integrated circuits combine many transistors on a single chip
- How to manufacture integrated circuits?
- <https://www.youtube.com/watch?v=bor0qLifjz4>
- Silicon: semiconductor
- Add materials to transform properties:
 - Conductors
 - Insulators
 - Switch

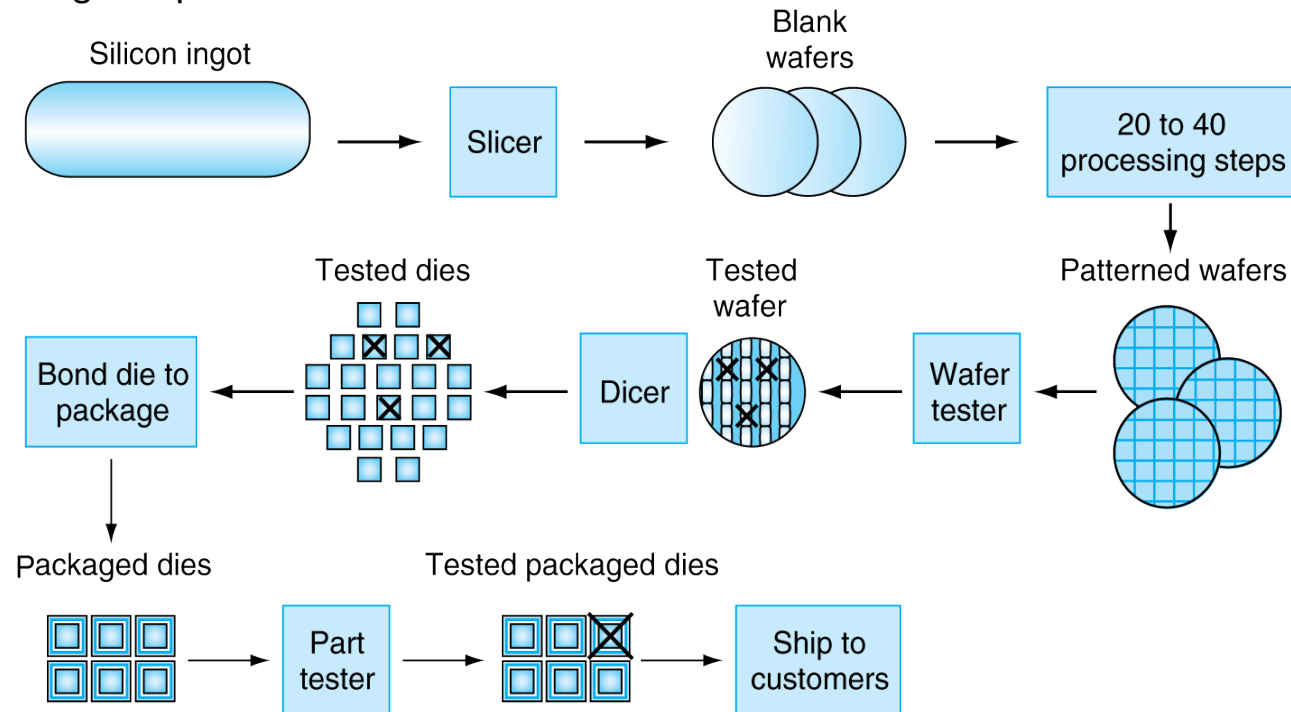
Intel Core i7 Wafer

300mm wafer, 280 chips, 32nm technology
Each chip is 20.7 x 10.5 mm



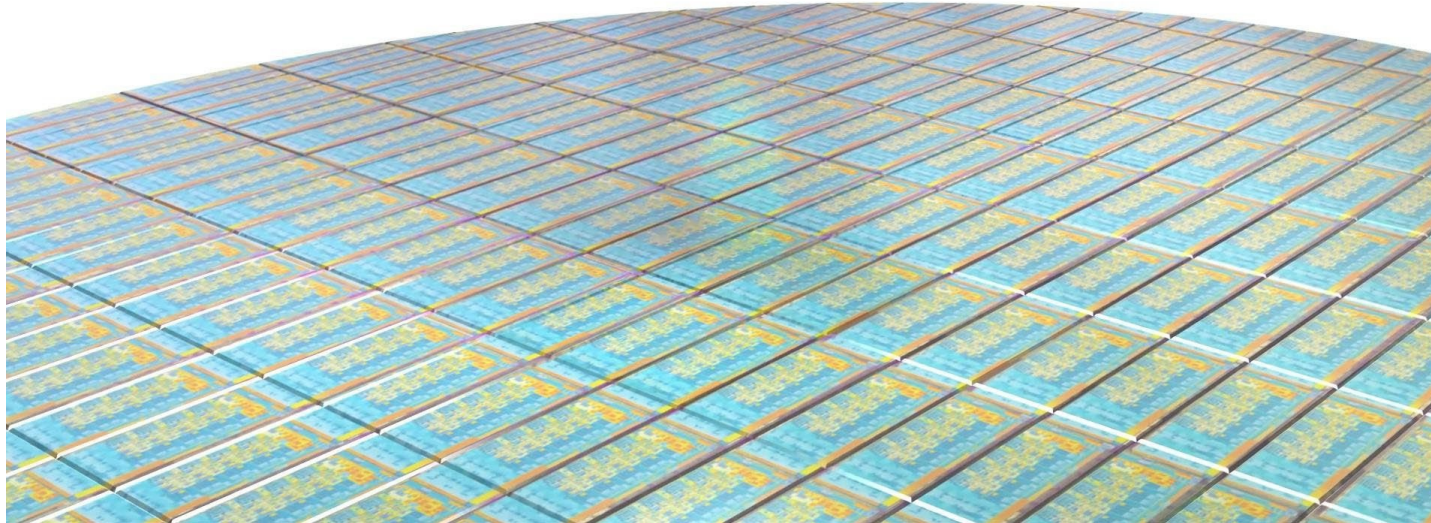
Manufacturing ICs

How to manufacture integrated circuits? Yield:
proportion of working dies per wafer



Smaller transistors allow more components per chip, improving performance and efficiency.

Intel: The Making of a Chip



<https://www.youtube.com/watch?v=d9SWNLZvA8g>

Explanations: http://download.intel.com/newsroom/kits/chipmaking/pdfs/Sand-to-Silicon_22nm-Version.pdf

1. Design for Moore's Law
2. Use abstraction to simplify design
3. Make the common case fast
4. Performance via parallelism
5. Performance via pipelining
6. Performance via prediction
7. Hierarchy of memories
8. Dependability via redundancy

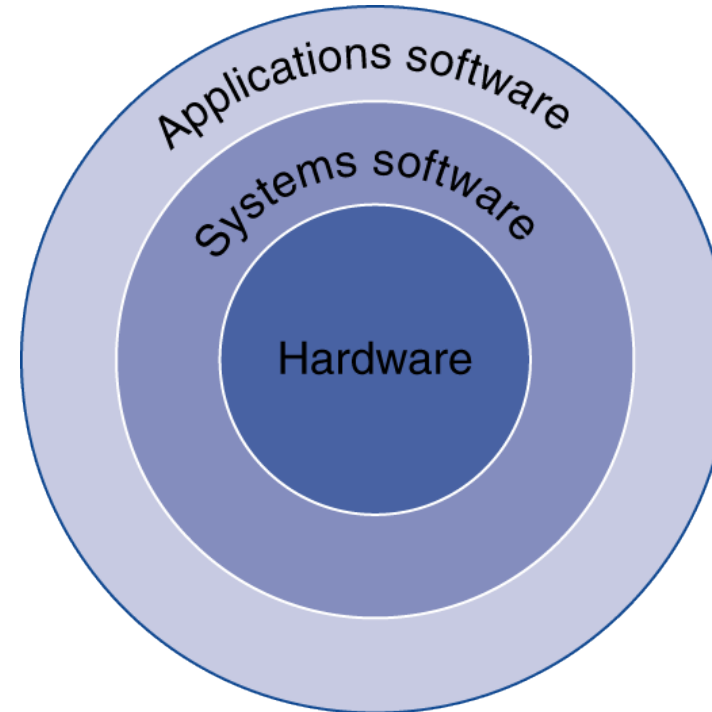


Eight Great Ideas in Computer Architecture



Components

- Application software
 - Written in high-level language
- System software
 - **Compiler: translates HLL code to machine code**
 - **Operating System: service code**
 - Handling input/output
 - Managing memory and storage
 - Scheduling tasks & sharing resources
- Hardware
 - Processor, memory, I/O controllers





System Software

The operating system acts as a bridge between applications and hardware.

The OS sits between hardware and user-level software (applications), providing:

- Isolation (e.g., assigning separate memory regions to processes))
- scheduling (e.g., managing CPU)
- Higher-level abstractions for low-level resources like IO devices.



Instruction Set



- An instruction set is a collection of operations or commands that a computer's central processing unit (CPU) can execute.
- The instruction set architecture (ISA) defines the interface between the hardware and the software, serving as a bridge that allows software programs to communicate with the CPU.
- The ISA defines what instructions a CPU understands.
- Different computers have different instruction sets, but with many aspects in common.
- Early computers had very simple instruction sets.



Instruction Set Architectures (ISAs)

- ISA defines the interface which hardware presents to software.
 - A compiler translates high-level source code (e.g., C,C++) to the ISA for a target processor.
 - The processor directly executes ISA instructions.

Example ISAs:

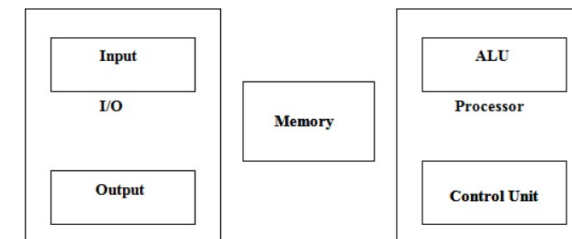
- MIPS (Microprocessor without Interlocked Pipeline Stages)
- ARM (popular on mobile devices)
- x86 (popular on desktops and laptops)



Instruction Set Architectures (ISAs)

Three basic types of instructions:

- Arithmetic/bitwise logic (ex: addition, left-shift, bitwise negation, xor).
- Data transfers to/from/between registers and memory.
- Control flow
 - Unconditionally jump to an address in memory
 - Jump to an address if a register has a value of 0
 - Invoke a function
 - Return from a function

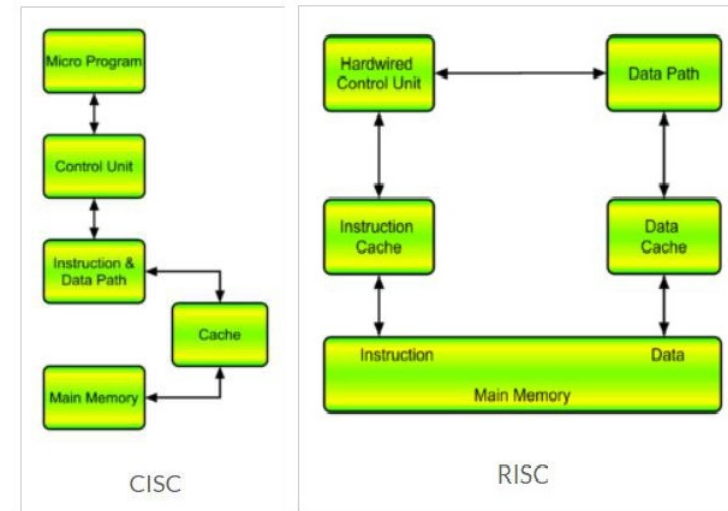


Functional units of computer

RISC vs.

CISC

- When processor manufacturer makes a processor, they decide which instruction set architecture (ISA) the processor speaks
- Two types:
 - **Reduced instruction set computing (RISC)**
 - very basic instructions, e.g. add, load, store
 - Example: MIPS
 - **Complex instruction set computing (CISC)**
 - several basic instructions combined into a single instruction
 - Example: x86

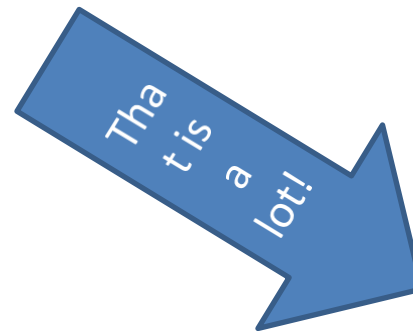


<https://www.ahirlabs.com/difference/cisc-vs-risc/>



RISC vs CISC: ISA Wars

- CISC (Complex Instruction Set Computer): ISA has a large number of complex instructions.
 - “Complex”: a single instruction can do many things.
 - Instructions are often variable-size to minimise RAM usage.
 - CISC instructions make life easier for compiler writers, but much more difficult for hardware designers—complex instructions are hard to implement.
 - **X86** is the classic CISC_ ISA



A single HW instruction has to do:

- A brunch
- A memory read
- A memory write
- A two register increment or decrement

RISC vs CISC: ISA Wars

- RISC (Reduced Instruction Set Computer):
ISA smaller number of simple instructions.
 - RISC hardware only needs to do a few, simple things
 - RISC ISAs make it easier to design fast, power- efficient hardware.
 - RISC ISAs usually have fixed-sized instructions and a load/store architecture.
 - Ex: MIPS, ARM



The MIPS Instruction

- The **MIPS (Microprocessor without Interlocked Pipeline Stages)** instruction set follows a **Reduced Instruction Set Computing (RISC)** architecture, widely utilised in various fields such as **embedded systems, workstations, and education.**
- **Dominates the embedded core market**
- **Used in consumer electronics, networking/storage devices, cameras, and printers**
- **Represents common features of modern Instruction Set Architectures (ISAs)**

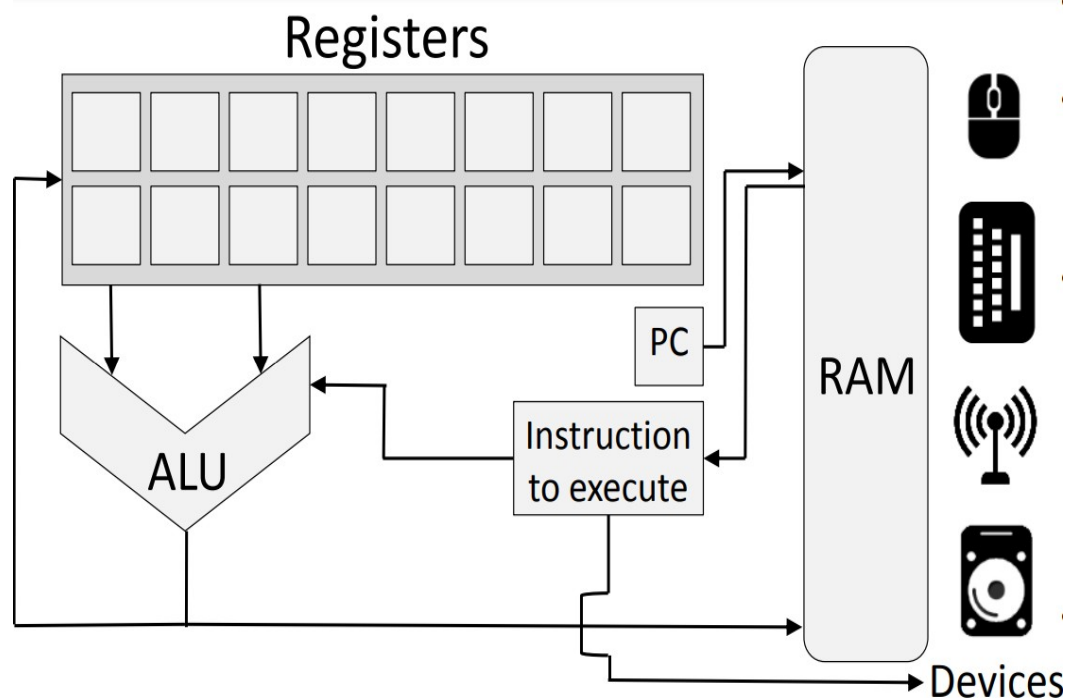


MIPS is widely used in education because it is simple and easy to understand.

Stanford MIPS commercialized by MIPS Technologies
(www.mips.com)

From the View of a Programmer

This view helps us understand how high-level programs are executed by hardware.



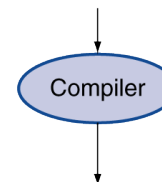
- Program instructions live in RAM
- PC register points to the memory address of the instruction to fetch and execute next
- Arithmetic logic unit (ALU) performs operations on registers, writes new values to registers or memory, generates outputs which determine whether a branches should be taken
- Some instructions cause devices to perform actions

Levels of Program Code

- High-Level Language
 - Offers abstraction close to the problem domain.
 - Enhances productivity and code portability.
- Assembly Language
 - Human-readable representation of machine instructions.
- Hardware Representation
 - Uses binary digits (bits) to encode instructions and data.

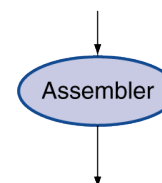
High-level
language
program
(in C)

```
swap(int v[], int k)
{int temp;
  temp = v[k];
  v[k] = v[k+1];
  v[k+1] = temp;
}
```



Assembly
language
program
(for MIPS)

```
swap:
  muli $2, $5, 4
  add $2, $4, $2
  lw $15, 0($2)
  lw $16, 4($2)
  sw $16, 0($2)
  sw $15, 4($2)
  jr $31
```



Binary machine
language
program
(for MIPS)

```
00000000101000010000000000011000
00000000000110000001100000100001
10001100011000100000000000000000
10001100111100100000000000000100
10101100111100100000000000000000
10101100011000100000000000000100
0000001111100000000000000001000
```

Each level hides complexity



Example Instructions: add, subtract

- Form of add and subtract instructions:
 - Three operands: two sources and one destination
`add a, b, c    # a gets b + c`
- All arithmetic instructions have this form
- **Design Principle:** Simplicity favours regularity
 - Regularity makes implementation simpler
 - Simplicity enables higher performance at lower cost



Example Instructions: add, subtract

C code:

```
f = (g + h) - (i + j);
```

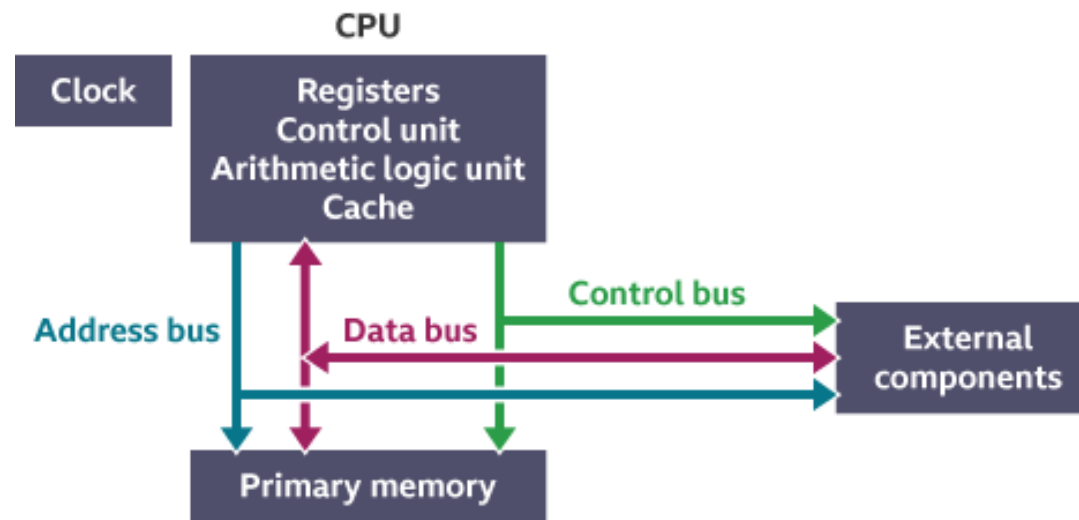
Compiled MIPS code:

```
add t0, g, h    # temp t0 = g + h
add t1, i, j    # temp t1 = i + j
sub f, t0, t1   # f = t0 - t1
```

This is simplified: we can't use variable names in MIPS!

The **Central Processing Unit (CPU)** consists of six main components:

- control unit (CU)
- arithmetic logic unit (ALU)
- registers
- cache
- buses
- clock



The CU provides several functions:

- it fetches, decodes and executes instructions
- it issues control signals that control hardware
- it moves data around the system

The ALU has two main functions:

- It performs arithmetic and logical operations (decisions). The ALU is where calculations are done and where decisions are made.
- It acts as a gateway between primary memory and secondary storage . Data transferred between them passes through the ALU.



Registers

Registers are small amounts of high-speed memory contained within the CPU. They are used by the processor to store small amounts of data that are needed during processing, such as:

- the address of the next instruction to be executed
- the current instruction being decoded
- the results of calculations



Thank You!